

UDP - User Datagram Protocol

Module 13

Duration: 22:53

Video: https://www.youtube.com/watch?v=g4p_kJxiRHk

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0:02	[Music]
0:15	Hello friends I am engineer Sunil Kumar from school of electrical engineering and information technology Punjab
0:22	Agriculture University Ludhiana I welcome you all to the video lecture series of cyber and information security
0:30	in this lecture we will study about UDP first of all please see the contents which I shall cover after covering brief
0:39	introduction about UDP I shall take up characteristics of user datagram which will be followed by UDP services in the
0:48	last I shall take up UDP features and UDP packages so now let us start UDP is
0:55	a simple transfer protocol that extends the host to host delivery of packets of the underlying Network into a process to
1:04	process communication since there are many processes running on a given host UDP needs to add a level of Dem
1:13	multiplexing which allowing multiple application processes on each host to share the network figure shows the
1:21	relationship of the user datagram protocol to the other protocols and layers of the tcpip protocol suit UTP is
1:29	located between the application layer and the IP layer and serve as the intermediary between the application
1:37	programs and the network operations a transport layer protocol usually has several responsibilities one is to
1:45	create a process to process communication UDP uses port number to accomplish this another responsibility
1:53	is to provide control mechanism at the transport layer UDP does this task at a very min minimal level there is no flow
2:02	control mechanism and there is no acknowledgement for received packets UDP does provide error control to some

2:09 extent if UDP detects an error in the received packet it silently drop it UTP

2:17 is a connectionless unreliable transfer protocol it does not add anything to the services of Ip except for providing

2:25 process to process communication instead of host to host communication UDP is a very simple protocol using a minimum of

2:34 overhead if a processor wants to send a small message and does not care about reliability it can use UDP sending a

2:43 small message using UDP takes much less interaction between the sender and the receiver than

2:50 TCP next user datagram UDP packets called user datagrams have a fixed size

2:57 header of 8 bytes figure shows the form of the user datagram the fields are as follows First Source port number this is

3:06 the port number used by the process running on the source host it is 16bit long which means that P number can range

3:16 from 0 to 65,535 if the source host is a client the port number in most cases is an inal

3:25 port number requested by the process and chosen by the UDP software running on the source host in the source host is

3:34 the server the p number in most cases is a well-known port number next destination port number this

3:42 is the port number used by the process running on the destination host it is also 16 bits long if the destination

3:50 host is the server the port number in most cases is a well-known port number if the destination host is the client

3:59 the port number in most cases and inferable port number in this case the server copies the inferable port number

4:06 it has received in the request packet this is a 16bit field that defines the total length of the user datagram and

4:14 header Plus data the 16 bits can define a total length of 0 to

4:21 65,535 bytes however the total length needs to be much less because a UDP user

4:28 datagram is stored in a and IP datagram with the total length of 65,535 bytes the length field is a UDP

4:37 user datagram is actually not necessary a user datagram is encapsulated in an IP datagram there is a field in IP datagram

4:47 that defines the total length there is another field in the IP datagram that defines the length of the header so if

4:55 we subtract the value of the second field from the first we can deduce the length of the user datagram that is

5:03 encapsulated in an IP datagram UTP length equal to IP length minus IP

5:11 headers length however the designers of the UDP protocol felt that it was more efficient for the destination UDP to

5:19 calculate the length of the data from the information provided in the UDP used datagram rather than ask the IP software

5:28 to supply this information we should remember that when the IP software delivers the UDP user data gram

5:35 to the UDP layer it has already dro the IP headers next checkm this field is used

5:43 to detect errors over the entire user datagram that is header Plus data I will discuss the dump example of UDP header

5:51 in hexadecimal format the UDP header is CB

5:57 8400 d 0 01 C 0 01 C some questions based on this UDP header first question

6:07 what is a source port number The Source port number is the first four hexadecimal digits like C B

6:15 846 which means that the source Board number is 5210 next question what is a destination

6:23 port number the answer is that the destination port number is the second four hexadecimal digits that are 0 0 0

6:32 D14 which means that the destination port number is 13 next question what is the total length of the user datagram

6:40 the third four hexadecimal digits 0 0 1 c16 defines the length of the whole UDP

6:47 packets as 28 bytes next question what is the length of data the length of the

6:54 data is the length of the whole packet minus the length of the header means 28 - 8 equal to 20 bytes next question is

7:03 the packet directed from a client to a server or visce Versa answer since the destination port number is 13 the packet

7:11 is from the client to the server next question what is the client process the client process is the

7:19 daytime next UDP Services I discuss the general Services provided by the UDP in

7:26 transport layer first process to process communication UDP provides process to process communication using sockets with

7:35 the combination of IP addresses and port numbers several port numbers used by UDP

7:41 are as follows port s is for Eco Port 9 is for discard port 11 is for users port

7:50 13 for daytime Port 17 is for COD Port 19 is for charging port 53 is for domain

7:58 Port 6 67 is for boot PS Port 68 is for boot PC Port 69 is for tftp Port 111 is

8:08 for RPC Port 123 is for ntp Port 161 is for SNMP Port 162 is for SNMP

8:18 trap next service is connectionless service UDP provides a connectionless service this means that each user

8:26 datagram sent by UTP is an independent datagram there is no relationship between

the different users datagram

- 8:34 even if they are coming from the same Source process and going to the same destination program the user datagrams
- 8:42 are not numbered also there is no connection establishment and no connection termination as is the case of
- 8:49 TCP this means that each user datagram can travel on a different path one of the ramifications of being
- 8:57 connectionless is that the processes that uses UDP cannot send a stream of
- 9:03 data to UDP and expect UDP to chop them into different related users datagrams
- 9:10 instead each request must be small enough to fit into one user datagram in the case of UDP only those processes
- 9:18 sending short messages when messages less than 6557 bytes next flow control UDP is a
- 9:27 very simple protocol there is no flow control and hence no window mechanism the receiver May overflow with incoming
- 9:35 messages the lack of flow control means that the process using UDP should provide for this service if it is needed
- 9:45 next error control there is no error control mechanism in a UDP except for the check sum this means that the sender
- 9:54 does not know if a message has been lost or duplicated when when the receiver detects an error through the check sum
- 10:02 the user datagram is silently discarded the lack of error control means that the process using UTP should provide for
- 10:11 this service if it is needed next checkm UDP check sum calculation is different
- 10:18 from the 104 IP here the checkm includes three sections a pursuer header a UDP
- 10:26 header and the data coming from the application layer the pursu header is the part of the header of the IP packet
- 10:33 in which the user datagram is to be encapsulated with some Fields filled with zeros if the check sum does not
- 10:42 include pedo header a user datagram may arrive safe and sound however if the IP
- 10:49 header is corrupted it may be delivered to the wrong host the protocol field is added to ensure that the packets belong
- 10:58 to UDP and to TCP we will see later that if a process can use either UDP or TCP
- 11:06 the destination p number can be the same the value of the protocol field for UDP is 17 if this value is changed during
- 11:15 the transmission the check some calculation at the receiver will detected and UDP drops the packet it is
- 11:23 not delivered to the wrong protocol note the similarities between the pursued header fields and the last 12 bytes of

11:31 the IP address next congestion control since UDP is a connectionless protocol it does not provide congestion control

11:40 UDP assumes that the packets sent are small and cannot create congestion in the network this assumption may or may

11:48 not be true today when UDP is used for realtime transfer of audio and video

11:55 next encapsulation and de encapsulation to send a message from one process to another process the UDP protocol

12:02 encapsulates and de encapsulates the messages first encapsulation when a process has a message to send through

12:11 UDP it means the message to UDP along with the pair of socket addresses and the length of data UDP receives the data

12:20 and adds the UDP header UDP then passes the user datagram to IP with the socket addresses IP adds its own head

12:29 using the value 17 in the protocol field which indicating that the data has come from the UDP protocol the IP datagram is then passed to the data link layer the

12:36 data link layer receives the IP datagram which adds its own header and passes it to the physical layer the physical layer

12:42 encodes the bits into electrical or Optical signals and send it to the remote machine next decapsulation

12:51 when the message arrives at the destination host the physical layer decodes the signal into bits and passes

12:59 it to the data link layer the data link layer uses the header to check the data if there is no error the header and the

13:06 trailer are dropped and the datagram is passed to IP the IP software does its own checking if there is no error the header is dropped and the user datagram is passed to UDP with the sender and the

13:15 receiver IP addresses UDP uses the check sum to check the entire user datagram if there is no error the header is dropped

13:22 and the application data along with the sender socket addresses is passed to the processes the sender socket address is

13:31 passed to the processes in case it needs to respond to the message received next queuing in UDP qes are associated with ports at the client side when a process

13:40 starts it request a port number from the operating system some implementations create both an incoming and outgoing

13:48 cues associated with each process other implementations create only an incoming queue associated with each process note

13:55 that even if a process wants to communicate with multiple processes it obtains only one p number and eventually

14:02 one outgoing and one incoming queue the cues opened by the clients are in most

cases identified by eeral P numbers the

14:37 cues function as long as the process is running when the process terminates the cues are destroyed the client processes

14:45 can send messages to the outgoing cues by using Source Board number specified in the request UDP removes the messages

14:54 one by one and after adding the UDP header deliver them to IP an outgoing queue can overflow if this

15:02 happens the operating system can ask the client processes to wait before sending any more messages next multiplexing and

15:11 Dem multiplexing in a host running a tcpip protocol suit there is only one UDP but possibly several processes that

15:20 may want to use this service of UDP to handle this situation UDP multiplexes

15:26 and D multiplexes first mul Lexing at the center side there may be several processes that need to send user

15:35 datagrams however there is only one UDP this is a many to one relationship and

15:42 requires multiplexing UDP accepts messages from different processes differentiated by their assigned P

15:50 numbers after adding the header UDP passes the user datagram to IP next d

15:58 multiplexing at the receiver side there is only one UDP however we may have many

16:04 processes that canes user datagrams this is one to many relationship and requires

16:11 Dem multiplexing UDP receives user datagrams from IP after error checking and dropping of the header UDP delivers

16:20 each message to the appropriate process based on the port numbers next comparison between UDP and generic

16:28 simple protocols we can compare UDP with the connectionless simple protocol the only difference is that the UDP provides

16:36 an optional check sum to detect corrupted packets at the receiver side if the check sum is added to the packet

16:44 the receiving UDP can check the packet and discard the packet if it is corrupted no feedback however is sent to

16:52 the center UDP is an example of the connectionless simple protocol with the exception of an optional checkm added to

17:01 17 minutes, 1 second packets for error detection next UDP features we briefly discuss some

17:08 features of UDP and their advantages and disadvantages First connectionless Service as we mentioned previously UDP

17:16 is a connectionless protocol each UDP packet is independent from other packets sent by the same application program

17:24 this feature can be considered as an advantage or disadvantage depending on the

application requirement it is an

17:32 advantage if for example a client application needs to send a short request to a server and to receive a

17:39 short response if the request and response can each fit in one single user datagram a connectionless service may be

17:47 preferable the overhead to establish and close the connection may be significant in this case in the connection oriented

17:55 service to achieve the above goal at least nine packets are exchanged between the client and the server in

18:03 connectionless service only two packets are exchanged the connectionless service provides less delay and the connection

18:10 oriented service creates more delay if delay is an important issue for the application the connection L service is

18:17 preferred next lack of error control UDP does not provide error control it

18:24 provides an unreliable service most applications accept reliable services from a transport layer protocol although a

18:34 reliable service is desirable it may have some side effects that are not acceptable to some applications when a

18:42 transport layer provides reliable Services if a part of the message is lost or corrupted it needs to resend

18:50 this means that the receiving transport layer cannot deliver that part of the application immediately there is an

18:57 uneven delay between different parts of the message delivered to the application layer some applications by Nature do not

19:05 even notice these uneven delays but for some they are very crucial next lack of

19:12 congestion control UDP does not provide congestion control however UDP does not create additional traffic in an error

19:21 prone Network TCP May resent a packet several times and thus contribute to the creation of the congestion therefore in

19:30 some cases lack of error control in UDP can be considered an advantage when congestion is a big issue next UDP

19:39 package to show how UDP handles the sending and receiving of UD package we present a simple version of UDP package

19:48 we can say that UDP package involve five components first control Block Table

19:55 input q a control block module an input module and a output module figure shows the five components and

20:05 their interactions first control Block Table in our package UDP has control

20:11 Block Table to keep track of the open ports each entry in this table has a minimum of four Fields the state which

20:21 can be free or in use the process ID the port number and the corresponding Q
20:27 number next next input cues our UDP package uses a set of input cues one for
20:34 each process in this design we do not use output cues next control block module
the control block module is
20:44 responsible for the management of the control block table when a process starts it
asks for a port number from
20:52 the operating system the operating system assigns well-known port numbers to
servers and imperial port numbers to
21:00 the client the processes passes the process ID and the port number to the control
block module to create an entry
21:09 in the table for the process the field for Q number has a value of zero note that we
have not include a strategy to
21:18 deal with the table that is full next input module the input module receives a
21:24 user datagram from the IP it searches the control Block Table to find an entry
21:31 having the same port number as this user datagram if the entry is found the module
uses the information in the entry
21:40 to NQ the data if the entry is not found it generates an icmp message next output
21:49 module the output module is responsible for creating and sending user
21:55 datagrams so friends this was all about UDP hope the concepts explained in this
22:01 22 minutes, 1 second lecture were understandable and helpful hope to see you in
the next lecture till then goodbye thank
22:10 [Music]
22:27 you for
22:34 [Music]